Pedro Boechat

Rugby, UK
pboechat@gmail.com
pedroboechat.com

linkedin
github
stackoverflow
videos

Summary of Qualifications

Software Engineer
- 18 years in the software industry (2 semi-conductor, 8 games, 8 enterprise)
- 4 years in academia

Computer Scientist
- 2 papers published in SIGGRAPH Asia

Indie Developer
- $15k prize winner in a nationwide game contest in Brazil

Work Experience

Staff Engineer — **Arm** — **Apr 2026 - Present**

Senior Graphics Engineer — **Arm** — **Dec 2023 - Apr 2026**

Senior Graphics Programmer — **nDreams** — **Jan 2023 - Nov 2023**
- Worked as a graphics specialist on an unannounced title for VR:
  - Optimized the rendering pipeline for improved efficiency.
  - Collaborated with artists to set best practices and performance budgets.
  - Conducted graphics experiments to measure performance and refine visual quality.
  - Assisted in updating the game across multiple engine versions (5.0, 5.1, and 5.3).

Lead Programmer — **Lab42 Games** — **Jan 2021 - Dec 2022**
- Ported Grid Legends from PC/high-end consoles to the Meta Quest 2:
  - Ported the codebase to Android.
  - Extended the engine's Vulkan renderer to support multiple features needed by a tile-based renderer.
  - Restructured the engine's forward rendering path.
  - Tweaked the engine's core systems to better conform to the platform's hardware capabilities.
  - Integrated OpenXR to the engine.
  - Extended the engine's asset building pipeline to support ASTC and OBB packing.

Work Experience

Senior Programmer — **Codemasters** — **Aug 2018 - Dec 2020**

- Worked on the new world editor used in the F1 game series:
  - Decoupled the entity system from the old editor and re-integrated it into the new editor.
  - Increased the robustness of the communication between the editor and the engine.
  - Extended the engine's entity system to support edit-time components.
  - Created a visual editor for the cutscene system scripting language.

Lead Programmer — **Bongfish** — **Apr 2017 - Jul 2018**

- Worked on the development of tools to synthesize realistic game scenery from real-world data for the new Microsoft Flight Simulator:
  - Used ML techniques to segment and classify data from aerial images.
  - Built a cloud-based system to run large scale processes.
  - Generated 3D models at run-time using procedural modeling techniques.

Senior Programmer — **Bongfish** — **Apr 2016 - Mar 2017**

- Worked on the Frontline game mode for World of Tanks:
  - Helped implementing new game mechanics, such as the respawn system.
  - Refactored important entities, such as the in-battle inventory.

Research Assistant — **Graz University of Technology** — **Apr 2014 - Mar 2016**

- Researched GPU programming techniques applied to procedural generation:
  - Co-authored 2 publications in Siggraph Asia (once as first author and another as collaborator).
  - Co-supervised 2 Master's students.
- Developed a procedural scene generator for Bongfish under grant program (Innovationscheck 2016).

Research Assistant — **Pontifical Catholic University** — **Mar 2012 - Mar 2014**

- Worked on a virtual training application for oil rig workers:
  - Implemented head-tracking and stereoscopy using 3rd party tools
  - Deployed the application on the largest digital CAVE in Brazil.

Lead Programmer — **Decatron** — **Jan 2012 - Feb 2012**

- Worked on an online learning platform for bus transport professionals.

Senior Programmer — **OWSE** — **Feb 2011 - Dec 2011**

- Worked on an online maritime logistics system in one of the busiest port terminals in Brazil.

Work Experience

Senior Programmer — Message IT — **Jan 2009 - Jan 2011**

- Developed the company's application framework and many of the company's internal tools.

Programmer — Message IT — **Jan 2007 - Dec 2008**

- Worked on an online billing system for Globosat, the biggest Pay-TV company in Brazil.

Programmer — Petrobrás Social Security Foundation — **Sep 2004 - Dec 2006**

- Worked in the migration of company's loan system to the web.

Education

Graz, Austria — **Graz University of Technology** — **Apr 2014**
Ph.D. in Computer Graphics (Not completed)

Rio de Janeiro, Brazil — **Pontifical Catholic University** — **Jan 2012 - Mar 2014**
M.Sc. in Informatics, Major in Computer Graphics (GPA: 9,1)

Rio de Janeiro, Brazil — **Centro Universitário da Cidade** — **Jan 2008 - Dec 2011**
B.Sc. in Computer Science (Merit scholarship)

Areas of Expertise

Graphics programming, Engine programming, C++, Python, OpenGL, Vulkan, GLSL, HLSL, CUDA, GPGPU, Game development, Software engineering, Test-driven development